

ORCA Replenishment Policy and Decision Framework

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PRIORITY RULE: This document contains planning rules and context only. All live values (costs, contacts, balances) must be fetched from orca.db via MCP tools. The database is the single source of truth for all facts.

Section 1 — ABC Class Ordering Rules

ORCA portfolio: 102 SKUs across 7 categories. ABC classification drives ordering rules, approval thresholds, and option eligibility. Always fetch abc_class from curated_skus via get_sku_info.

Class	# SKUs	Characteristics	Ordering Rules	Option B?	Lead Buffer
A (High Value)	~20	High margin, high velocity Critical to customer experience	Min 200 units regardless of shortfall Full distribution — all affected stores Weekly review mandatory Stockout triggers VP notification	NEVER Full distribution required	+20% on effective lead time
B (Medium)	~37	Medium margin and velocity Standard portfolio SKUs	Follow supplier min_order_qty All affected stores (unless HIGH pressure) Bi-weekly review	Permitted (Flagged if margin rank > 5)	+10% recommended
C (Low Value)	~45	Low margin, low velocity Consolidate where possible	Standard only — no expedite unless Tier-1 stockout Batch with same supplier May exclude during HIGH pool pressure	Permitted	None required

Section 2 — Stock Status Response Rules

Stock status computed from days_of_cover vs effective_lead_time. Always use the value in curated_inventory.stock_status — never recompute.

Status	Threshold	Action Required	Urgency	SLA
CRITICAL	days_of_cover < 50% of effective_lead_time OR stock = 0	Immediate action Evaluate expedite first All Critical stores included	CRITICAL if >5 stores or lead_time_too_late HIGH otherwise	24 hours
AT RISK	days_of_cover between 50% and 100% of effective_lead_time	Standard replenishment Acceptable unless event uplift At Risk may be excluded from Option B	HIGH if any Critical exists MEDIUM otherwise	48 hours
HEALTHY	days_of_cover ≥ 100% of effective_lead_time	No action required Weekly monitoring	N/A	None
OVERSTOCK	current_stock > 3× reorder_point	Flag for markdown or inter-store transfer Use CP006 Store Transfer Reserve	N/A	None

Section 3 — Capital Allocation Scoring Formula (Agent 3)

Every feasible option is scored 0–100. Highest score wins. Rules applied in strict order — elimination before scoring.

Step	Rule / Formula	Max Points	Notes
RULE 1 Elimination	If option.pool_pressure_flag = HIGH → feasible = False	100	Applied FIRST before any scoring. HIGH pressure pool = option eliminated.
RULE 2 Elimination	If total_cost_aed > pool.available_aed → feasible = False	100	Budget check. Cannot exceed live available balance.
RULE 3 Flag	If abc_class = A AND option.id = B → not_recommended = True	100	Class A needs full distribution. Option B (Tier-1 only) flagged, not eliminated.

Step	Rule / Formula	Max Points	Notes
RULE 4a Budget Score	$\text{budget_score} = (1 - \text{cost} \div \text{available_budget}) \times 40$	40 pts	Lower cost relative to budget = higher score. Option B cheapest → highest budget_score.
RULE 4a Availability Score	$\text{availability_score} = \text{availability_pct} \times 0.40$	40 pts	More stores covered = higher score. Option A (all stores) scores highest here.
RULE 4a Margin Score	$\text{margin_score} = (1 \div \text{margin_priority_rank}) \times 20$	20 pts	Rank 1 (highest margin) = 20 pts. Rank 5 = 4 pts. Rank 10 = 2 pts.
RULE 4b Penalty	If urgency = CRITICAL AND lead_time_days > 30: total_score -= 20	-20 pts	CRITICAL urgency only. Penalises slow options during critical events. Option A with 54-day lead gets -20 if CRITICAL.
RULE 5 Approval	$\text{approval_required} = (\text{cost} > \text{pool.auto_approve_limit_aed})$		Fetch auto_approve_limit from database. True = human approval needed via HITL.

Section 4 — Option Building Rules (Agent 2)

Option	Name	Order Qty Formula	Lead Time	Pool	Stores Served
A	Standard Replenishment	$\text{MAX}(\text{min_order_qty}, \text{projected_shortfall})$	effective_lead_time	CP001 or CP004	All Critical and At Risk
B	Profit Maximisation	$\text{MAX}(\text{min_order_qty}, \text{ROUND}(\text{projected_shortfall} \times 0.6))$	effective_lead_time	CP001 or CP004	Tier-1 stores only ■ Not for Class A
C	Expedite Air Freight	$\text{MAX}(\text{min_order_qty}, \text{projected_shortfall}) \times (1 + \text{expedite_premium}/100)$	effective_lead_time × 0.35	CP003 only	All stores Critical first ■ Requires allows_expedite=Yes